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|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|-------------|
| Programme      | BSc Mathematics Honours                                                                                                                                                                                                 |                           |                    |             |
| Course Code    | MAT4CJ203                                                                                                                                                                                                               |                           |                    |             |
| Course Title   | <b>REAL ANALYSIS I</b>                                                                                                                                                                                                  |                           |                    |             |
| Type of Course | <b>Major</b>                                                                                                                                                                                                            |                           |                    |             |
| Semester       | IV                                                                                                                                                                                                                      |                           |                    |             |
| Academic Level | 200 – 299                                                                                                                                                                                                               |                           |                    |             |
| Course Details | Credit                                                                                                                                                                                                                  | Lecture/Tutorial per week | Practicum per week | Total Hours |
|                | 4                                                                                                                                                                                                                       | 3                         | 2                  | 75          |
| Pre-requisites | 1. Mathematical Logic and necessary exposure to set theory.<br>2. Basic Calculus                                                                                                                                        |                           |                    |             |
| Course Summary | After introducing the basic notions in set theory, the course develops into the construction of the Real number system. Thereafter Real functions are introduced and the notions of limit and continuity are developed. |                           |                    |             |

#### Course Outcomes (CO):

| CO                                                                                                                                                                                             | CO Statement                                                                                                                                                                                                                                                   | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|-----------------------------------------------------------------------|
| CO1                                                                                                                                                                                            | Demonstrate Proficiency in Set Theory Fundamentals and Real Number Properties                                                                                                                                                                                  | An               | C                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/Report/<br>End Sem Exam |
| CO2                                                                                                                                                                                            | Apply the completeness property of $\mathbb{R}$ , and solve problems involving intervals and applications of the supremum property.                                                                                                                            | U                | C                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/Report/<br>End Sem Exam |
| CO3                                                                                                                                                                                            | Analyse sequences and their limits, apply limit theorems, and demonstrate an understanding of concepts such as monotone sequences, sub-sequences, and the Cauchy Criterion, as well as their applications in solving problems related to sequences and limits. | An               | C                   | Internal Exam/<br>Assignment/Seminar/<br>Viva/Report/<br>End Sem Exam |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                                                                                                                                                |                  |                     |                                                                       |

**Detailed Syllabus:**

| Textbook | Introduction to Real Analysis, 4/e, Robert G Bartle, Donald R Sherbert John Wiley & Sons (2011) |                                                                                                                                                                                          |                |                           |
|----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|
| Module   | Unit                                                                                            | Content                                                                                                                                                                                  | Hrs<br>(45+30) | External<br>Marks<br>(70) |
| I        | Introduction to Set theory                                                                      |                                                                                                                                                                                          | 8              | Min.15                    |
|          | 1                                                                                               | Section 1.1 - Sets and functions (for review only)                                                                                                                                       |                |                           |
|          | 2                                                                                               | Section 1.2 - Mathematical Induction (Proofs of results included in practicum part).                                                                                                     |                |                           |
|          | 3                                                                                               | Section 1.3 – Finite and Infinite sets.                                                                                                                                                  |                |                           |
|          | 4                                                                                               | Section 1.3 – Countable and Uncountable sets.                                                                                                                                            |                |                           |
| II       | The Real numbers                                                                                |                                                                                                                                                                                          | 13             | Min.15                    |
|          | 5                                                                                               | Section 2.1 – The algebraic properties of $\mathbb{R}$ .                                                                                                                                 |                |                           |
|          | 6                                                                                               | Section 2.1 – The order properties of $\mathbb{R}$ .                                                                                                                                     |                |                           |
|          | 7                                                                                               | Section 2.2 – Absolute value and the Real Line.                                                                                                                                          |                |                           |
|          | 8                                                                                               | Section 2.3 – Completeness property of $\mathbb{R}$ (Proofs included in Practicum).                                                                                                      |                |                           |
|          | 9                                                                                               | Section 2.4 – Applications of the Supremum property - 2.4.3 to 2.4.6 and 2.4.8 to 2.4.9 (All other discussions included in Practicum).                                                   |                |                           |
|          | 10                                                                                              | Section 2.5 – Intervals – 2.5.2 to 2.5.4 (All other discussions included in Practicum).                                                                                                  |                |                           |
| III      | Sequences and Limits                                                                            |                                                                                                                                                                                          | 12             | Min.15                    |
|          | 11                                                                                              | Section 3.1 – Sequences and their limits.                                                                                                                                                |                |                           |
|          | 12                                                                                              | Section 3.1 – Problems to find limits of sequence.                                                                                                                                       |                |                           |
|          | 13                                                                                              | Section 3.2 – Limit theorems.                                                                                                                                                            |                |                           |
|          | 14                                                                                              | Section 3.2 – Problems using Limit theorems.                                                                                                                                             |                |                           |
|          | 15                                                                                              | Section 3.3 – Monotone sequences – Monotone Convergence Theorem.                                                                                                                         |                |                           |
|          | 16                                                                                              | Section 3.3 – Applications of Monotone Convergence Theorem – Euler’s number introduction only.                                                                                           |                |                           |
| IV       | Sequences and Limits (continued)                                                                |                                                                                                                                                                                          | 12             | Min.10                    |
|          | 17                                                                                              | Section 3.4 – Sub sequences and the Bolzano Weierstrass theorem (Second proof of Theorem 3.4.8 is omitted for external exam and limits superior and inferior are included in practicum). |                |                           |
|          | 18                                                                                              | Section 3.4 – Problems using Divergence criteria.                                                                                                                                        |                |                           |
|          | 19                                                                                              | Section 3.5 – The Cauchy Criterion (Examples 3.5.9, 3.5.11 and Corollary 3.5.10 are included in Practicum).                                                                              |                |                           |
|          | 20                                                                                              | Section 4.1- Limits of functions (Proofs included in Practicum).                                                                                                                         |                |                           |
|          | 21                                                                                              | Section 4.2: Limit theorems of functions (Proofs included in Practicum).                                                                                                                 |                |                           |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------|----------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 22                                                                                                                                                                                                                                                                                                                | Section 4.3: Some extensions of limit concepts (Proofs included in Practicum).                                         |           |          |
| <b>V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Practicum:</b><br>The goal is for the students to learn the following topics in 15 practicum sessions of two hours each via self-study and group activities. The lecturer may assist by running group discussions, supervising class seminars and referring library books for self-study and note preparation. |                                                                                                                        | <b>30</b> | <b>-</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                 | Section 1.2 - for detailed discussions including proofs                                                                |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2                                                                                                                                                                                                                                                                                                                 | Section 2.3 – re do it with all the proofs                                                                             |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3                                                                                                                                                                                                                                                                                                                 | Section 2.4 – Worked out examples for applying the ideas of supremum and infimum and the existence of square root of 2 |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4                                                                                                                                                                                                                                                                                                                 | Section 2.5 – Characterization theorem for intervals and representations of real numbers                               |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 5                                                                                                                                                                                                                                                                                                                 | Section 3.4 – discussions of limit inferior and limit superior with examples                                           |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 6                                                                                                                                                                                                                                                                                                                 | Section 3.5 – Estimation of errors in contractive sequences with examples                                              |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7                                                                                                                                                                                                                                                                                                                 | Section 3.6 – Properly divergent Sequences                                                                             |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8                                                                                                                                                                                                                                                                                                                 | Section 3.7 – Introduction to Infinite Series – conditions for convergence – Harmonic Series                           |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9                                                                                                                                                                                                                                                                                                                 | Section 3.7 – Comparison Tests with examples                                                                           |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10                                                                                                                                                                                                                                                                                                                | Section 4.1 – Formulate a precise definition of limit and illustrate with examples                                     |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 11                                                                                                                                                                                                                                                                                                                | Section 4.1 – Sequential Criterion for Limits for convergence and divergence with examples                             |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 12                                                                                                                                                                                                                                                                                                                | Section 4.2 – Limit theorems for functions in parallel to that of sequences.                                           |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 13                                                                                                                                                                                                                                                                                                                | Section 4.3 – One sided and infinite limits.                                                                           |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 14                                                                                                                                                                                                                                                                                                                | Section 11.1 – Open sets, their properties and characterization.                                                       |           |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 15                                                                                                                                                                                                                                                                                                                | Section 11.1 - Closed sets, their properties and characterization.                                                     |           |          |
| <b>References</b> <ol style="list-style-type: none"> <li>1. Tom.M. Apostol, Calculus I, Wiley &amp; Sons.</li> <li>2. Tom.M. Apostol, Mathematical Analysis, 2/e, Addison-Wesley.</li> <li>3. Richard R Goldberg, Methods of Real Analysis, 2/e, Wiley</li> <li>4. Raymond L Wilder, Introduction to the Foundations of Mathematics,2/e, John WileySons</li> </ol>                                                                               |                                                                                                                                                                                                                                                                                                                   |                                                                                                                        |           |          |
| <b>Optional Programming References for Practicum:</b> <ol style="list-style-type: none"> <li>(1) SageMath Calculus Tutorial <a href="https://www.sagemath.org/calctut/limits.html">https://www.sagemath.org/calctut/limits.html</a></li> <li>(2) SageMath 2D plotting <a href="https://doc.sagemath.org/html/en/reference/plotting/sage/plot/plot.html#">https://doc.sagemath.org/html/en/reference/plotting/sage/plot/plot.html#</a></li> </ol> |                                                                                                                                                                                                                                                                                                                   |                                                                                                                        |           |          |

**\*70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

**Mapping of COs with PSOs and POs:**

|      | PSO1 | PSO2 | PSO3 | PSO4 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|------|------|------|------|------|-----|-----|-----|-----|-----|-----|-----|
| CO 1 | 2    | 1    | 3    | 2    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |
| CO 2 | 1    | 3    | 2    | 2    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |
| CO 3 | 3    | 2    | 3    | 3    | 3   | 0   | 3   | 0   | 3   | 0   | 0   |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics:**

|      | Internal Exam | Assignment/ Report | Seminar | Viva | End Semester Examinations |
|------|---------------|--------------------|---------|------|---------------------------|
| CO 1 | ✓             | ✓                  | ✓       | ✓    | ✓                         |
| CO 2 | ✓             | ✓                  | ✓       | ✓    | ✓                         |
| CO 3 | ✓             | ✓                  | ✓       | ✓    | ✓                         |